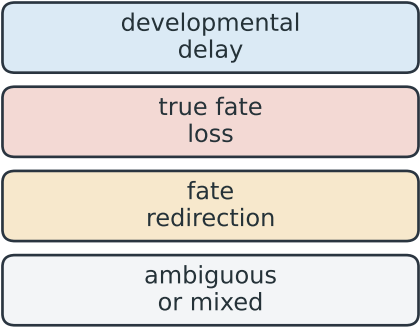


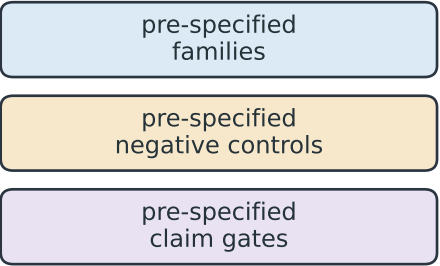
Figure 4. Real-data generalization through event modes and locked validation

A. Full TED event-mode grammar



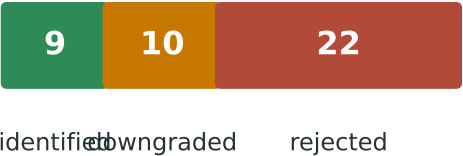
ZSCAPE: embryo blocks + real time + perturbations

B. Locked adapter validation



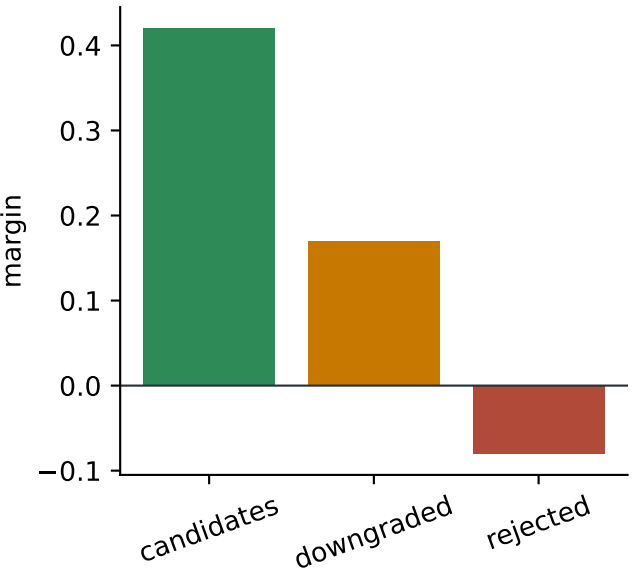
GSE157977 was not used to tune TED parameters.

C. Guide-level outcomes

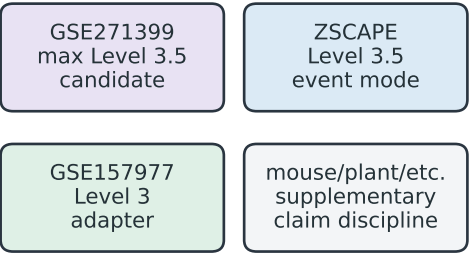


41 guide barcodes; ceiling remains Level 3 adapter evidence.

D. Negative-control margin



E. Evidence-level preservation



F. Main-text rule

Generalization is not defined as forcing every dataset into full TED.

Lower-evidence systems remain supplementary claim-discipline examples.

No cell-type fate loss or functional causality is claimed without required gates.